



Effect of artificial intelligence on banking stability: Evidence from developed countries

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ABSTRACT

This article investigates the effect of artificial intelligence on banking stability in developed countries. The study uses panel data from 29 developed countries over the period 2017 to 2021. To analyze the data, a threshold model combined with quantile regression and the local projection method was employed. The results show that artificial intelligence positively influences banking stability in developed countries. However, this effect depends on the level of banking regulation: when the capital adequacy ratio reaches a critical threshold of 7.95%, the impact of artificial intelligence on banking stability becomes significantly stronger. Moreover, this effect varies but remains robust across different quantiles of banking stability within financial institutions. Finally, using the local projection method, the results reveal that a one-time shock in investment in artificial intelligence leads to a statistically significant positive effect on banking stability three years after the shock. In conclusion, this article offers economic policy recommendations aimed at encouraging investment in artificial intelligence to strengthen banking stability in developed countries.

1. Introduction

The development of new technologies based on artificial intelligence (AI) is generating significant debate in the literature, with several authors questioning its potential impact on economies. Initial analyses suggest that the adoption of AI-related technologies fundamentally alters labor market dynamics, often leading to disruptions (Acemoglu et al., 2022; Webb, 2020; Yilmazkuday, 2024). These labor market disruptions are not without implications for the financial sector. Banks, in particular, are extensively leveraging machine learning technologies to manage wage payments and address the needs of credit applicants. Scott et al. (2024) highlight that machine learning models enable more accurate predictions of default probabilities, allowing for the implementation of proactive strategies to mitigate risk. These models also help to minimize the risk associated with asymmetric information, one of the factors driving credit market imbalances (Stiglitz and Weiss, 1981).

Additionally, Danielsson et al. (2022), through a synthesis of the literature, examine how AI is transforming the operations of the financial system. Similarly, Atadoga et al. (2024) explore the opportunities for innovation and efficiency that AI presents for the banking sector in the United States.

Artificial intelligence (AI) is used to predict phenomena such as risk

and requires large amounts of data to learn and improve. It can help financial institutions identify the riskiest customers, or alert banks to the need to monitor certain profiles more closely. Financial institutions are therefore forced to adopt this innovation to avoid, in the long term, disappearing from the financial market. For example, the advance of AI in the banking sector is also helping to optimize processes and improve the customer experience (Sträßer & Stolicna, 2023). According to Königstorfer and Thalmann (2020), AI reduces loan losses, enhances payment security, automates compliance tasks and refines customer targeting all of which boost overall performance.

Other studies identify various transmission channels through which artificial intelligence can affect the banking system and its stability. Among them, the operational efficiency channel. Through this channel, AI enables process automation, cost reduction and productivity enhancement, strengthening banks' resilience (Petràlia et al., 2019). Mensah et al. (2024) point out that artificial intelligence and machine learning can increase efficiency, accuracy and customer experience in the banking sector. However, these technologies also pose significant challenges, such as discrimination, algorithmic biases, loss of privacy and the need to strengthen cybersecurity.

AI can also enhance bank stability through the risk management channel, in which financial institutions mobilize advanced algorithms such as machine learning and deep learning to improve their ability to

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detect and manage credit, fraud and bank liquidity risks (Zhang & Zhou, 2019; Fuster et al., 2022). These institutions exploit large volumes of data, including unstructured data from alternative sources, to better predict default probabilities. This advanced analysis capability enables finer segmentation of borrowers according to their risk profile and financial behavior. As a result, credit granting decisions are more targeted towards creditworthy customers, which in turn helps to reinforce the stability of financial institutions.

In recent years, global investment in AI and related technologies has surged, reaching \$150 billion by 2022, with over 75% originating from developed countries (Adrian, 2024). These investments are primarily directed toward automating banking processes, detecting fraud, and optimizing risk management systems. Banks in advanced economies, such as those in the European Union, the United States, and Japan, have been integrating these technologies extensively into their operations. A 2023 survey conducted by the European Central Bank (ECB) reveals that 85% of major European banks now use AI for risk management. According to the Wi-FiTalents report published in 2025¹, statistics on the adoption of artificial intelligence vary depending on the area of application. The report indicates that 52% of banks have already integrated AI into their customer service operations, and 44% are using AI-powered chatbots to interact with customers.

In the marketing sector, over 60% of banks use AI to run personalized campaigns. Additionally, 72% reported improved customer satisfaction after adopting AI. The report also reveals that 62% of banks use AI algorithms such as machine learning and deep learning to perform personalized risk assessments. In this context, fraud detection systems now identify 73% of fraud cases, leading to a 35% reduction in fraud-related losses thanks to AI-based anomaly detection. It is therefore clear that intelligent algorithms are not only shaping the future they are already reinforcing the financial systems of today.

Despite the growing adoption of artificial intelligence, empirical studies investigating its link to the stability of financial institutions remain limited (Basri & Almutairi, 2023; Amina et al., 2023). This study aims to address this gap by analyzing the effect of artificial intelligence on the stability of banks in developed countries. The main hypothesis is that the adoption of artificial intelligence (AI) contributes to enhancing banking stability in the economies under study. However, this effect is conditional on the level of banking regulation. Specifically, when banking regulation measured by the capital adequacy ratio is below a certain threshold, the adoption of AI has a positive but limited effect on banking stability. In contrast, when regulation exceeds this threshold, the impact of AI becomes significantly stronger. This is because stricter regulation ensures better control over credit allocation to economic agents. In such a regulatory environment, AI can be more effectively leveraged to analyze borrower profiles, predict default risks, and support banks in granting loans to the most creditworthy clients. As a result, the quality of loan portfolios improves, thereby reinforcing the overall stability of the financial system.

The primary contribution of this research lies in exploring how machine learning technologies can enhance bank stability in developed countries. This analysis provides valuable insights for financial institutions, enabling them to better manage risks and accelerate their stabilization processes, thereby reducing the risk of exclusion from financial systems.

The remainder of this paper is structured as follows: Section 1 introduces the study, while Sections 2 and 3 discuss the theoretical and empirical approaches, respectively. Section 4 presents the methodology and results. Finally, the conclusion outlines the implications for economic policy.

2. Literature review

In recent years, economic literature has shown increasing interest in the impact of artificial intelligence (AI) on global development. This literature review focuses on the relationship between the adoption of AI and key economic development indicators, particularly in the banking sector.

Several studies have identified factors that influence banking stability (Sonehekpon, 2024; Sodokin et al., 2023; Triki et al., 2017). Regarding AI, its large-scale implementation such as the use of advanced systems like OpenAI can enable banks to enhance efficiency, convenience, and security (Dybek, 2023). Other studies also highlight that the adoption of green banking technologies such as chatbots, facial recognition, and fraud detection significantly and positively enhances bank stability (Basri & Almutairi, 2023; Amina et al., 2023), as well as banks' responsiveness to climate change initiatives. Moreover, it partially and positively mediates the relationship between AI-based green banking technologies and bank stability (Amina et al., 2023). Perals et al. (2025) reveal in their studies a positive relationship between private investment in AI and the efficiency of financial institutions for a sample of 24 countries.

Beyond this relationship between banking stability and artificial intelligence, several other studies highlight a link between AI and bank performance, showing that AI adoption positively influences both bank financial performance (Alzaidi, 2018; Elegunde & Shotunde, 2020; Iqbal & Fikri, 2025; Basri & Almutairi, 2023) and customer satisfaction, particularly in Pakistan (Iqbal & Fikri, 2025). Naeemi et al. (2024) also show that, when bank performance is measured using indicators such as net profit margin, return on equity (ROE) and return on assets (ROA), investment in artificial intelligence contributes positively to this performance. Concerning Jordanian banks, Shiyyab et al. (2023) prove, using data from 2014 to 2021, that AI significantly improves financial performance by boosting both ROA and ROE. Gyau et al. (2024) confirm this analysis by showing how AI can improve return on assets (ROA), considered an indicator of banking performance. In Chinese banks, AI adoption exerts a stronger influence on sustainable environmental performance than digital banking alone. This also demonstrates that AI mediates between external pressures and sustainability outcomes (Siddik et al., 2025). Following on from other work, AI and machine learning are shown to increase efficiency, accuracy and customer experience in the banking sector. However, these technologies also pose significant challenges, such as discrimination, algorithmic biases, loss of privacy and the need to strengthen cybersecurity (Basri & Almutairi, 2023; Mensah et al., 2024).

On the other side, Dybek(2023) emphasizes that banks must prioritize mobile banking solutions to better meet customer needs. Danielsson et al. (2022) provide a synthesis of the existing literature on how AI can transform financial systems. They demonstrate that AI has the potential to improve risk management and facilitate interactions with financial authorities. However, they also caution that AI could introduce vulnerabilities by creating new risks or amplifying existing ones. These risks include procyclicality, heightened uncertainties, increased reliance on trust, and optimizations that may inadvertently undermine the stability of the system.

In line with Naim's (2022) study, which employs an in-depth qualitative analysis, AI has been shown to benefit both service providers and users. These findings have been corroborated in financial companies, particularly through detailed analyses of balance sheets and annual reports, which reveal a generally positive perception of AI adoption. Using data from the banking sector in Turkey, Demirel and Topcu (2024) demonstrate that the implementation of AI fosters the growth of digital banking. Similarly, Atadoga et al. (2024) highlight the significant opportunities AI creates for innovation and efficiency in the U.S. banking sector. Furthermore, Elshamly et al. (2023) emphasize that adopting AI-based green banking technologies such as chatbots, facial recognition, and fraud detection substantially enhances banks' stability while

¹ Wi-FiTalents report published in 2025, statistics on the adoption of artificial intelligence

supporting climate change initiatives.

In the context of risk management within financial institutions, AI has proven effective in mitigating various risks, including credit risk, operational risk, liquidity risk, and reputational risk (Swankie and Broby, 2019). This finding is further supported by Scott et al. (2024), who demonstrate that AI-based machine learning models can predict default probabilities with high accuracy, thereby enabling the development of proactive risk mitigation strategies within banking institutions. Another fundamental contribution from the literature emphasizes that AI can reduce loan losses, enhance payment security, automate compliance tasks, and refine customer targeting, all of which boost banks' overall performance (Königstorfer & Thalmann, 2020).

Conversely, in the labor market, AI appears to have adverse effects. Webb (2020) explores how AI accelerates the decline in employment and wages. Similarly, Acemoglu et al. (2022) illustrate that AI is fundamentally reshaping labor dynamics. Using data from Google Trends, Yilmazkuday (2024) reveals that AI is disrupting the job market. Employing the local projection method, Yilmazkuday shows that positive shocks to search trends related to "artificial intelligence" exacerbate labor market conditions at the state level, with the effects persisting for one month.

The observation in the literature reveals that the majority of works are mainly concerned with the synthesis of ideas rather than quantitative analyses to confirm the relationships between artificial intelligence and the banking sector, in particular. This study aims to fill this gap by examining the relationship between artificial intelligence and bank stability in developed countries.

3. Methodology

3.1. Variables and data sources

The data used in this study mainly come from the World Bank's World Development Indicators (WDI) for economic and financial variables, and from the Stanford Institute for Human-Centered Artificial Intelligence (HAI) for data on AI investments and the AI adoption index. The data cover the period from 2017 to 2021, based on the availability of information on AI investment and the AI index, which is calculated as the weighted average of all pillar scores (Fattorini et al., 2024).

For example, Peralis et al. (2025) used AI investment as a measure of AI adoption to evaluate its effect on banking efficiency in 24 countries. Similarly, Tingbani et al. (2025) and Giwa & Ngepah (2024) analyzed, respectively, how companies invest in AI and how AI investment affects low-skilled employment in OECD countries. In this context, the AI adoption index developed by Fattorini et al. (2024) adds to the literature by being tested on banking stability in the countries covered by this study. In addition, the Z-SCORE index is used to measure bank stability. This dependent variable is calculated using data from the World Development Indicators (WDI). Originally introduced by Roy (1952), the index is calculated as the sum of the return on assets and the ratio of total equity to total assets, divided by the standard deviation of the return on assets. It reflects the likelihood of default in a country's banking system. Other control variables such as the capital adequacy ratio (CAR), GDP growth rate, inflation rate (INFLA), and agricultural value added were selected based on the existing literature on factors that may influence banking stability (Sonehekpon, 2024; Sodokin et al., 2023; Triki et al., 2017; Giwa & Ngepah, 2024; Peralis et al., 2025). Table 1 below presents the variables included in the study, along with their definitions and sources.

3.2. Model

The aim of this paper is to assess the effect of artificial intelligence (AI) on the stability of banks in developed Countries, with the goal of formulating policy recommendations that could inform AI investment strategies on a global scale.

Table 1

Variables and data sources.

Variables	Descriptions	Sources
Z-SCORE	The instability index, developed by Roy (1952), is calculated as the sum of the return on assets and the ratio of total equity to total assets, divided by the standard deviation of the return on assets. This index reflects the probability of failure within a country's banking system.	WDI
AI_INVEST	Private investment in artificial intelligence per capita (in billions of US dollars).	HAI
AI_Index_V	The AI vibrancy index for a country k , denoted as V_k , is calculated as the weighted average of the scores of all the pillars (Fattorini et al., 2024)	HAI
CAR	Capital adequacy ratio ("equity / total assets"), an indicator of banking regulation.	WDI
GDP	Economic growth rate.	WDI
INFLA	Inflation rate.	WDI
VA	Added value in the agricultural sector.	WDI

Source: Author's elaboration

First, I establish the relationship between bank stability and AI adoption by examining bank behavior in the context of credit-granting activities. Let us consider a representative bank whose objective is to maximize its profit $B(X)$, from granting X in credit. The bank's profit can be expressed as $B(X) = rX - C(X)$, where r is the interest rate. I assume that $C(X)$, representing the total cost of accessing credit, is composed of three components: the cost of accessing credit, the cost of using artificial intelligence (AI), and the cost of a unit of banking stability.

Assuming Z-score is the bank's stability and AI is the variable measuring the use of AI, I postulate the total cost function related to the credit granting activity in the following form $C(X) = \theta X^2 + \alpha.X.(Z - score) + \beta \frac{1}{AI(X)}$.

The first part θX^2 represents the cost of granting credit, which increases quadratically. As credit volume rises, the cost increases at an accelerating rate. The second part $(\alpha.X.(Z - score))$ reflects how bank stability influences the cost of credit. When a bank is more stable, it is likely to incur a higher cost in a context of fixed interest rates. In fact, bank stability can play a complex role in the cost of accessing credit. When a bank is stable, it may become more selective in choosing borrowers, which could lead to an increase in the cost of accessing fixed-rate credit. This is because the bank seeks to offset potential risks and maintain profitability, thus driving up the cost of credit for certain borrowers.

The third part $\frac{1}{AI(X)}$, measures the cost of managing default risk or assessing customers' ability to repay using artificial intelligence (AI). AI is employed to predict the risks associated with X . The extensive use of AI reduces the cost of risk management. The β coefficient is used to adjust this relationship. In this context, the bank's profit maximization program can be expressed as follows:

$$\max_X B(X) = \left[rX - \left(\theta X^2 + \alpha.X.(Z - score) + \beta \frac{1}{AI(X)} \right) \right] \quad (1)$$

The first-order condition gives the following equation:

$$\frac{\partial B(X)}{\partial X} = r - 2\theta X - \alpha.(Z - score) + \beta \frac{(AI(X))'}{[AI(X)]^2} = 0 \quad (2)$$

From this Eq. (2), we can obtain the following relationship:

$$Z - score = \frac{1}{\alpha} \left[r - 2\theta X + \beta \frac{(AI(X))'}{[AI(X)]^2} \right] \quad (3)$$

Relationship (3) assumes that the Z-score is a function of the interest rate r , credit X , and AI usage. It suggests that bank stability exhibits a non-linear relationship with AI usage. Assuming Z represents other characteristics that influence bank stability, the Z-score function can be expressed as follows:

$$Z - \text{score} = f(r, X, AI, Z) \quad (4)$$

Assuming that this existing relationship between banking stability and the other variables is dynamic, we can rewrite the Z-score function in the following form:

$$Z - \text{score}_{it} = g(r_{it}, X_{it}, AI_{it}, Z_{it}) \quad (5)$$

Given that the relationship between stability appears to be non-linear, I propose a panel threshold model to econometrically test the existence of a non-linear relationship between artificial intelligence and banking stability. To this end, I employ the Panel Threshold Regression (PTR) model proposed by Hansen (1999) to estimate this non-linear effect. The PTR model, developed by Hansen (1999), uses the fixed-effects estimator and introduces a threshold at which a regime change may occur. It helps identify whether banking regulation, measured by the capital adequacy ratio (CAR), significantly alters the relationship between AI and banking stability. The structural equation of this model is formulated as follows:

$$Y_{it} = u_i + \beta_1' X_{it} I(q_{it} \leq \delta) + \beta_2' X_{it} I(q_{it} > \delta) + \epsilon_{it} \quad (6)$$

Where $I(\cdot)$ is the indicator function. δ represents the threshold to be determined, X_{it} represents the model characteristics and q_{it} means the threshold variable. In our model, q_{it} represents banking regulation (CAR). I have determined the necessary banking regulation threshold at which the effect of artificial intelligence on banking stability changes. I also test the significance of the banking regulation threshold using the test proposed by Hansen (1999). With regard to this structural equation, the empirical specification can be written in the following form:

$$z - \text{score}_{it} = u_i + \beta_1' X_{it} I(CAR_{it} \leq \delta) + \beta_2' X_{it} I(CAR_{it} > \delta) + \epsilon_{it} \quad (7)$$

Where $z - \text{score}_{it}$ indicates the instability index developed by Roy (1952). It is calculated as the sum of the return on assets and the ratio of total equity to total assets, divided by the standard deviation of the return on assets. It reflects the probability of failure of a country's banking system. The Z-score compares the buffer of a country's banking system (capitalization and returns) with the volatility of these returns. When this indicator decreases, it indicates higher risk. Its formula is as follows:

$$Z - \text{score} = \left(\frac{ROA + \text{Equity}/\text{Total assets}}{\sigma(ROA)} \right) \quad (8)$$

Where ROA represents the return on banking assets and $\sigma(ROA)$ the standard deviation of the return on banking assets.

Before estimating the non-linear econometric relationship, I first employed both the fixed-effects and random-effects models to test the relationship between artificial intelligence and bank stability. To choose between the fixed-effects and random-effects models, I conducted the Hausman (1978) test. The null hypothesis is H_0 : the random-effects model, while the alternative hypothesis is H_1 : the fixed-effects model. The decision rule for this test is as follows: if the P-value is less than 5%, I reject the null hypothesis in favor of the fixed-effects model.

In addition, I use the "Global Vibrant Tools" index developed by Fattorini et al. (2024) alongside investment in artificial intelligence for the robustness analysis of our result. The GVT index uses various indicators from the fields of research and development, economics or responsible artificial intelligence, which are really organized into distinct pillars grouped within a global index. This relaunch introduces new AI dynamism sub-indices, including the Innovation Index, the Economic Competitiveness Index and the Policy, Governance and Public Engagement, and Public Engagement Index (Fattorini et al., 2024). The GVT is designed to adapt more effectively to the rapidly changing AI landscape than static indices. The calculated index first leads to the min-max normalization of all the indicators considered, after which the pillar j score for a country k is calculated. This score is calculated as the weighted average of various indicators associated with the pillar and the country. Assuming now p_{jk} the pillar score j for country k , the score is given by the following mathematical formula:

$$p_{jk} = \frac{\sum_{i=1}^{N_j} (w_{ij} \cdot x_{ijk})}{\sum_{i=1}^{N_j} w_{ij}} \quad (9)$$

Where N_j is the number of indicators for pillar j , x_{ijk} is the normalized score of the i th indicator for pillar j and country k and w_{ij} is the weight assigned to the i th indicator for pillar j , ranging from 0 to 10, with higher values indicating greater importance of the indicator. Fattorini et al. (2024) assume that these weights are non-negative, and their sum is non-zero.

For this purpose, the AI vibrancy index for a country k , denoted as V_k , is calculated as the weighted average of the scores of all the pillars as follows:

$$V_k = \frac{\sum_{j=1}^M (w_j \cdot p_{jk})}{\sum_{j=1}^M w_j} \quad (10)$$

Where M is the total number of pillars and p_{jk} is the score of pillar j for country k , as calculated above. w_j is the weight assigned to pillar j , ranging from 0 to 10, with higher values indicating greater importance of the pillar. Fattorini et al. (2024) assume that these weights are non-negative, and their sum is non-zero.

Finally, I focus on fixed effects, random effects, panel threshold regression, and panel quantile regression approaches to compare the robustness of the results, particularly because the sample of countries is small and the study period is very short. The advantage of fixed effects is that they control for unobserved heterogeneity by allowing each cross-sectional unit to have its own intercept, thus reducing bias from time-invariant omitted variables. Their main limitation is that they do not allow the estimation of variables that do not vary over time.

At the same time, random effects exploit both within- and between-variation, but their limitation lies in the risk of bias if the assumption of no correlation between unobserved effects and explanatory variables is violated. According to Machado & Silva (2019), panel quantile regressions make it possible to analyze the heterogeneous effects of explanatory variables at different points of the conditional distribution of the dependent variable, rather than only on the mean in our case, the different levels of bank stability. The panel threshold model of Hansen (1999) allows us, in this context, to capture the threshold of banking regulation beyond which the effect of AI on banking stability may switch regimes, provided that sufficient data are available.

4. Results and discussion

4.1. Descriptive statistics for variables

Table 2 summarizes the descriptive statistics for the variables used in this study. The statistics indicate that the average banking stability index (Z-SCORE) for developed countries is 18.68, with significant variation ranging from a minimum of 6.86 to a maximum of 39.34. These results reflect notable differences in the resilience of banking systems across these economies. As Danielsson and Uthemann (2023) emphasize, even in developed countries, financial systems can be vulnerable if regulatory and technological infrastructures are insufficient.

On average, developed countries invest US\$17.28 billion in AI, with a maximum investment of US\$257.35 billion. This disparity highlights the strategic importance of AI in certain advanced economies, while others lag behind. This finding is consistent with the observations of Maple et al. (2023), who stress the need for targeted investments to fully leverage the benefits of AI in the banking sector.

Fig. 1 presents a comparison between investment in artificial intelligence (AI_INVEST) and banking stability, as measured by the Z-SCORE, across several countries. The data reveal significant disparities. For instance, countries like Israel and the USA exhibit particularly high levels of AI investment, while others, such as Finland and Portugal, show much more modest levels. In contrast, the Z-SCORE, which gauges

Table 2
Descriptive statistics.

Variables	Observations	Mean	Standard deviation	Skewness	Kurtosis	Min	Max
Z-SCORE	145	18.68	7.82	0.90	3.07	6.86	39.34
AI_INVEST	145	17.28	34.06	4.03	23.25	0	257.35
AI_Index_V	145	18.27	17.57	1.72	5.81	0.15	81.85
GDP	145	2.29	4.11	-0.57	4.50	-11.16	15.12
INFLA	145	2.22	2.84	3.64	18.97	-1.138	19.59
CAR	145	8.08	1.98	0.85	3.48	4.86	14.85
VA	145	2.78	3.36	2.90	12.15	0.030	18.66

Source : author's calculation; **Footnote:** The high kurtosis of AI_INVEST, indicating non-normality. However, this does not affect the quality of the estimators of the models estimated below.

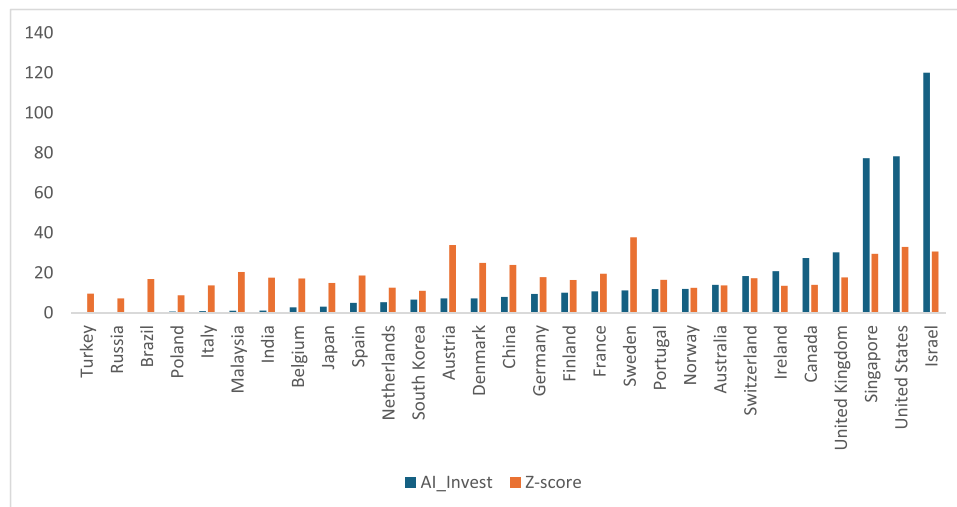


Fig. 1. Graph of investment in Artificial Intelligence (AI) and banking stability (Z-SCORE) by country (Source: created by the author).

banking stability, is generally more homogeneous, though notable variations exist between countries.

These results suggest a potential correlation between high AI investment and enhanced banking stability, as observed in countries like the USA and Singapore, where both indicators are relatively high. However, this relationship appears weak. For example, Israel stands out for its exceptionally high AI investment but relatively low Z-SCORE, indicating that substantial investment in technology does not automatically correlate with banking system stability. Similarly, countries such

as Sweden and Austria, whose banking systems are among the most stable in the sample, demonstrate relatively low levels of AI investment.

Furthermore, Fig. 2 illustrates the evolution of average investment in artificial intelligence (AI), the AI vibrancy index, and the bank stability index between 2017 and 2021. The analysis presented in this figure examines the relationship between AI investment and bank stability. Fig. 2 shows that AI investments have experienced remarkable growth, rising from US\$6.75 billion in 2017 to US\$37.45 billion in 2021, representing more than a fivefold increase over just five years. This trend

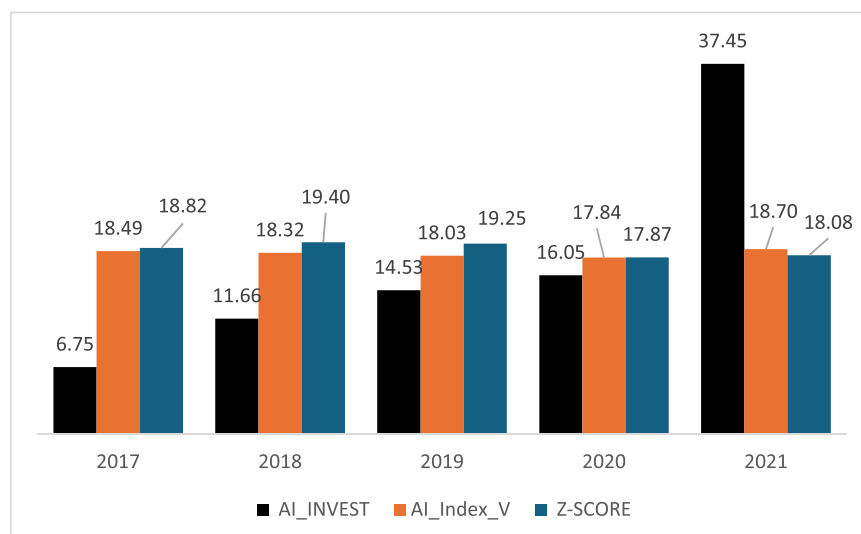


Fig. 2. Average investment in Artificial Intelligence (AI) and banking stability (Z-SCORE) between 2017-2021.

highlights a growing interest in AI, driven by its disruptive potential across various sectors, including finance.

In contrast, the banking stability index follows a less consistent trajectory. After a slight increase in 2018 (19.4), it gradually declined from 2019 onwards, reaching 17.87 in 2020, before rising slightly to 18.08 in 2021. These fluctuations can be attributed to external factors, such as global economic or health crises, particularly the COVID-19 pandemic, which directly impacted the stability of banking systems. Despite the surge in AI investment, banking stability appears to be influenced more by other structural and economic factors, suggesting that AI has not yet played an immediate and significant role in enhancing banking stability on a global scale.

4.2. Estimating the effect of AI investment on bank stability

Table 3 presents the econometric results for the effect of investment in artificial intelligence on banking stability in developed countries. Column 1 shows the model estimate using OLS, Column 2 presents the estimate using the fixed-effects (FE) model, and Column 3 uses random effects (RE). Column (4) presents the estimation results using the first-difference GMM estimator developed by Arellano and Bond (1991), aimed at addressing the endogeneity issue arising from potential reverse causality. This issue may occur, for instance, in a country where regulatory frameworks already promote banking stability, which in turn could encourage banks to increase their investments in artificial intelligence. The first-difference GMM estimator is used due to the relatively short time period covered by the study. This estimator employs both level and differenced instruments to generate more robust results compared to the fixed effects estimator, which only addresses bias related to time-invariant endogeneity in the estimations.

For the sample considered, the Hausman test statistic is 5.07. The corresponding p-value (0.280) exceeds the conventional thresholds of 1%, 5%, and 10%. As a result, the null hypothesis of no correlation between individual effects and explanatory variables is not rejected. Therefore, I prefer to adopt the random-effects model and use the GCM estimator. The random-effects model is deemed the most suitable for estimating the effect of artificial intelligence on banking stability in developed countries. In this model, the R-squared value of 63.89% indicates that 63.89% of the variations in banking stability in developed

countries are explained by the model's explanatory variables.

Considering the results from the econometric estimates presented in Table 3 of the random-effects model, investment in AI has a positive and significant effect on banking stability, at a 1% threshold. A 1% increase in AI investment leads to an 8.78% increase in banking stability.

Inflation (INFLA), on the other hand, has a negative and significant effect, indicating that inflationary pressures harm banking stability, even in advanced economies. This finding aligns with Sonehekpon (2024) conclusions about the harmful effects of inflation on bank stability. This result can be interpreted as a consequence of ineffective monetary policies, which create growing uncertainty that negatively affects banking regulation and, consequently, bank stability. Other variables, such as GDP and agricultural value added (VA), do not show significant effects in these models, which may reflect indirect or contextual dynamics. Moreover, the results from the first-difference GMM estimator in column (4) confirm the effect of AI on bank stability, indicating the robustness of our estimations.

4.3. Estimation of the PTR model for the effect of artificial intelligence investment on banking stability

The results from Table 4, derived from the estimation of the threshold model (PTR), highlight a nonlinear relationship between investment in artificial intelligence (AI) and banking stability. The critical banking regulation threshold, measured by the capital adequacy ratio (CAR), is estimated at 7.95%. The analysis shows that the effect of AI on banking stability varies significantly depending on this threshold, illustrating the importance of a robust regulatory framework to maximize the benefits of AI in the financial systems of developed countries.

In the first regime ($CAR \leq 7.95\%$), the effect of AI investment on banking stability is positive but relatively moderate. This can be explained by structural limitations in banks operating under a weak regulatory framework. In such conditions, the potential benefits of AI, such as improved risk management or anomaly detection, are hindered by organizational inefficiencies and inadequate supervision. This finding aligns with the conclusions of Abbas et al. (2024), who highlight the challenges faced by financial institutions operating in lightly regulated environments.

In the second regime ($CAR > 7.95\%$), the effect of AI investment becomes significantly more substantial. This reflects the fact that better-regulated banking systems provide an environment conducive to the integration and exploitation of advanced technologies, such as AI. These results align with the work of Chen et al. (2024), who demonstrated that institutions with strict regulation are able to fully leverage technological innovations to strengthen their resilience and improve operational efficiency.

Table 3
Estimation of the effect of AI on banking stability.

Variables	(1) OLS	(2) FE	(3) RE	(4) First diff GMM
Lag of Z-SCORE (t-1)	-	-	-	-0.064 (0.077)
Lag of Z-SCORE (t-2)	-	-	-	-0.062 (0.084)
AI_INVEST	0.087*** (0.018)	0.101*** (0.019)	0.0878*** (0.018)	0.103*** (0.018)
CAR	0.024 (0.324)	0.016 (0.348)	0.024 (0.324)	-0.089 (0.318)
GDP	-0.022 (0.156)	-0.202 (0.267)	-0.022 (0.156)	-0.112 (0.248)
INFLA	-0.665*** (0.237)	-0.612** (0.242)	-0.665*** (0.237)	-0.559** (0.242)
VA	0.114 (0.197)	0.169 (0.199)	0.114 (0.197)	0.243 (0.181)
Constant	18.18*** (2.519)	18.16*** (2.567)	18.18*** (2.519)	20.34 (3.53)
Significance	7.36***	8.22***	36.81***	56.24***
R ²	0.209	0.233	0.6389	-
Observations	145	145	145	130
Hausman Test		5.07(0.280)		
Sargan test	-	-		130.04 (0.314)

Standard errors in parentheses: *** p<0.01, ** p<0.05, * p<0.1

Source: created by the author. Notes: RE=Random effect and FE=Fixed Effect

Table 4
Estimation of the nonlinear effect of AI_INVEST on banking stability.

Variables	Regime 1	Regime 2
	$\delta = 7.95^{***}$	
	$CAR_{it} \leq \delta$	$CAR_{it} > \delta$
AI_INVEST	0.0616*** (0.0230)	0.177*** (0.0281)
CAR	2.057*** (0.656)	0.781* (0.450)
GDP	0.0920 (0.314)	-0.184 (0.291)
INFLA	0.334 (1.084)	-0.610*** (0.229)
VA	-0.686** (0.340)	0.771*** (0.233)
Constant	5.810 (4.215)	
Significance	7.58***	
Observations	145	
R ²	0.368	

Standard errors in parentheses: *** p<0.01, ** p<0.05, * p<0.1

Regarding the other variables, banking regulation (CAR) has a positive and significant effect in both regimes. However, this effect is more pronounced in the first regime, which can be attributed to its central role in stabilizing poorly regulated systems. Furthermore, inflation (INFLA) has a negative and significant effect only in the second regime. This suggests that even in well-regulated environments, inflationary pressures can disrupt the financial system by increasing economic uncertainties and making it more difficult to manage banking risks. Finally, agricultural value added (VA) shows contrasting effects depending on the regime. In the first regime, its effect is negative, likely due to the high uncertainties in the agricultural sector, which increase borrower default risks. In the second regime, the effect becomes positive and significant, suggesting that a strong regulatory framework can mitigate sectoral risks and enhance banking stability, even in uncertain economic contexts.

4.4. Estimation of the effect of AI investment on banking stability using quantile regression

Table 5 analyzes the effect of investment in AI on banking stability across different quantiles of the conditional distribution of banking stability. This approach captures the heterogeneity of the effects, revealing how banks with varying levels of stability respond to investments in AI. The results show that the effect of AI investment is positive and significant across all quantiles (10th, 25th, 50th, 75th, and 90th). However, this effect is particularly pronounced for the 50th and 75th quantiles, corresponding to banks with intermediate or slightly higher stability. This may be explained by the fact that these institutions, which already have a solid foundation but are not yet fully optimized, benefit the most from the efficiency and resilience gains brought by AI. This finding aligns with the observations of Gomber et al. (2017), who emphasize the crucial role of advanced technologies in enhancing the performance of banks with an appropriate organizational structure.

In the lower quantiles (10th and 25th), the effect of AI remains positive but relatively less significant. This can be explained by structural or operational constraints in these banks, which limit their ability to fully integrate AI technologies. For example, inadequate technological infrastructure or a lack of specialized skills may reduce the potential benefits of AI in these institutions. These findings align with the work of Maple et al. (2023), who emphasize that the effectiveness of AI depends on the technological and organizational maturity of banks.

In the higher quantiles (90th quantile), the effect of AI is also less pronounced, although it remains significant. This may indicate that the most stable banks have already reached a certain level of optimization and gain less marginal benefit from new AI investments. Additionally, these banks may focus more on other strategies, such as international expansion or service diversification, to maintain their competitiveness.

Inflation (INFLA) has a negative and significant effect primarily in the intermediate quantiles (50th and 75th). This result confirms that

macroeconomic uncertainty linked to inflationary pressures affects bank stability, even in advanced economies. This aligns with the findings of Sonehekpon (2024), who suggests that inflation is harmful to bank stability. Finally, agricultural value-added (VA) has a positive and significant effect in the lower quantiles but becomes less significant in the higher quantiles. This can be explained by the fact that banks with lower stability are more reliant on their exposure to primary sectors, such as agriculture, for their financial resilience.

4.5. Robustness check: effect of the AI index developed by Fattorini et al. (2024) on banking stability

This section (Table 6) presents the robustness results concerning the effect of artificial intelligence on banking stability, using the AI index V developed by Fattorini et al. (2024). The results show that the effect of AI on banking stability remains robust for all model specifications when this index is used. The impact of artificial intelligence, as measured by the index V, is positive and significant at the 1% threshold for both the fixed-effects and random-effects models. This effect remains positive in both regimes of the threshold effects model. Furthermore, the regulatory threshold at which the effect of artificial intelligence becomes more pronounced is almost identical to that observed in the case of AI investment. These results suggest that the application of artificial intelligence in the banking system helps strengthen the resilience of banks in the face of shocks, thereby enhancing their stability.

4.6. Robustness check: effect of the AI_Index_V developed by Fattorini et al. (2024) on banking stability using quantile regression

This section analyzes the effect of the artificial intelligence index (AI_Index_V) on banking stability using quantile regression. The results presented in Table 7 show that the artificial intelligence index has a significantly more pronounced positive effect on the lower quantiles than on the upper quantiles of the conditional distribution of banking stability in developed countries. This result remains robust when using the estimation with artificial intelligence investment and suggests that artificial intelligence has a heterogeneous effect across different levels of banking stability in developed countries. This can be explained by the diversification of the economies in the developed countries of our sample.

4.7. Response of bank stability to artificial intelligence shock

Table 8 presents the results of the impulse response of banking stability to a shock from artificial intelligence over a 3-year horizon, using the local projection method. Column (1) shows the results for AI Investment, while column (2) presents the results for the Artificial Intelligence Index (AI_Index_V). The results of the AI investment estimation suggest that a one-unit shock in AI investment is insignificant for

Table 5
Estimation of the effect of AI_INVEST on banking stability using quantile regression.

Variables	(1) 10th quantile	(2) 25th quantile	(3) 50th quantile	(4) 75th quantile	(5) 90th quantile	(6) 95th quantile
AI_INVEST	0.099*** (0.026)	0.099*** (0.020)	0.101*** (0.018)	0.102*** (0.029)	0.103* (0.059)	0.105 (0.088)
CAR	0.023 (0.431)	0.021 (0.334)	0.017 (0.300)	0.013 (0.472)	0.0052 (0.946)	-0.002 (1.411)
GDP	-0.337 (0.318)	-0.295 (0.247)	-0.232 (0.222)	-0.152 (0.349)	-0.009 (0.700)	0.119 (1.043)
INFLA	-0.457** (0.216)	-0.505*** (0.167)	-0.578*** (0.151)	-0.670*** (0.238)	-0.835* (0.476)	-0.984 (0.707)
VA	0.436*** (0.161)	0.354*** (0.125)	0.228** (0.114)	0.069 (0.183)	-0.213 (0.364)	-0.470 (0.529)
Observations	145	145	145	145	145	145

Standard errors in parentheses: *** p<0.01, ** p<0.05, * p<0.1

Table 6
Effect of AI index V on banking stability.

Variables	(1) OLS	(2) FE	(3) RE	(4)First diff GMM	(5) Threshold model	
					Regime 1	Regime 2
					$\delta = 7.949^{***}$	
					$CAR_{it} \leq \delta$	$CAR_{it} > \delta$
Lag of Z-SCORE (t-1)				-0.112 (0.070)		
Lag of Z-SCORE (t-2)				-0.066 (0.078)		
AI_Index_V	0.218*** (0.037)	0.224*** (0.038)	0.218*** (0.0377)	0.247*** (0.033)	0.155*** (0.051)	0.350*** (0.049)
CAR	-0.075 (0.315)	0.023 (0.340)	-0.075 (0.315)	-0.131 (0.298)	2.982*** (0.670)	1.067** (0.432)
GDP	0.009 (0.150)	-0.231 (0.262)	0.0097 (0.150)	-0.162 (0.232)	-0.043 (0.305)	-0.105 (0.282)
INFLA	-0.445* (0.235)	-0.458* (0.241)	-0.445* (0.235)	-0.373* (0.230)	0.564 (1.048)	-0.355 (0.229)
VA	0.348* (0.201)	0.398* (0.206)	0.348* (0.201)	0.510*** (0.179)	-0.52 (0.335)	1.146*** (0.245)
Constant	15.31*** (2.528)	14.85*** (2.612)	15.31*** (2.528)	17.93*** (3.36)	-2.676 (4.282)	
Significance	9.66***	9.75***	48.29***	81.39***	9.16***	
Obs	145	145	145	130	145	
R ²	0.258	0.265	0.2579	-	0.413	
Sargan test				130.50 (0.304)		

Standard errors in parentheses: *** p<0.01, ** p<0.05, * p<0.1

Table 7
Effect of AI_Index_V on banking stability.

Variables	(1) 10th quantile	(2) 25th quantile	(3) 50th quantile	(4) 75th quantile	(5) 90th quantile	(6) 95th quantile
AI_Index_V	0.263*** (0.088)	0.254*** (0.076)	0.229*** (0.052)	0.208*** (0.050)	0.170* (0.088)	0.131 (0.145)
CAR	-0.057 (0.867)	-0.037 (0.747)	0.012 (0.508)	0.055 (0.487)	0.131 (0.854)	0.211 (1.409)
GDP	-0.293 (0.644)	-0.278 (0.555)	-0.239 (0.377)	-0.206 (0.362)	-0.147 (0.634)	-0.085 (1.045)
INFLA	-0.214 (0.417)	-0.273 (0.360)	-0.426* (0.246)	-0.556** (0.239)	-0.786* (0.422)	-1.030 (0.684)
VA	0.739** (0.319)	0.657** (0.276)	0.443** (0.191)	0.261 (0.190)	-0.062 (0.340)	-0.404 (0.534)
Observations	145	145	145	145	145	145

Standard errors in parentheses: *** p<0.01, ** p<0.05, * p<0.1

Table 8
Response of bank stability to artificial intelligence shock.

Horizons	(1) AI Investment	(2) Artificial intelligence index (AI_Index_V)
At the Time of the Shock	-0.0004	0.009
One Year After the Shock	-0.021	0.057
Two Years After the Shock	-0.009	-0.010
Three Years After the Shock	0.067***	-0.018

Significance, *** p<0.01, ** p<0.05, * p<0.1

banking stability at the time of the shock, one year after the shock, and two years after the shock (Fig. 3). However, the cumulative impulses become statistically positive and significant starting three years after the shock. Regarding the AI index, the results of the estimation for the Artificial Intelligence Index suggest that a one-unit shock in AI investment is insignificant for banking stability over the entire three-year horizon. However, artificial intelligence, through its investment,

positively and significantly affects banking stability three years after the shock.

Conclusion

This article analyzes the effect of investment in artificial intelligence on banking stability in developed countries. The econometric tools used show that investment in artificial intelligence promotes better banking stability in developed countries. However, this impact strongly depends on macroprudential policy levels, measured by the capital adequacy ratio (CAR), highlighting the central role of public policies and appropriate regulatory frameworks. This effect is non-linear, as there is a minimum threshold of banking regulation beyond which the impact of investment becomes even more significant. These observations are corroborated by the estimation of the threshold model (PTR), which identifies a critical banking regulation threshold at 7.95%. Moreover, the results also reveal heterogeneity in the effect of artificial intelligence investment on the conditional distributions of banking stability in developed countries. Banks with intermediate stability benefit the most from AI investments, while those at the extremes (very stable or very fragile) show more moderate responses. However, the study has

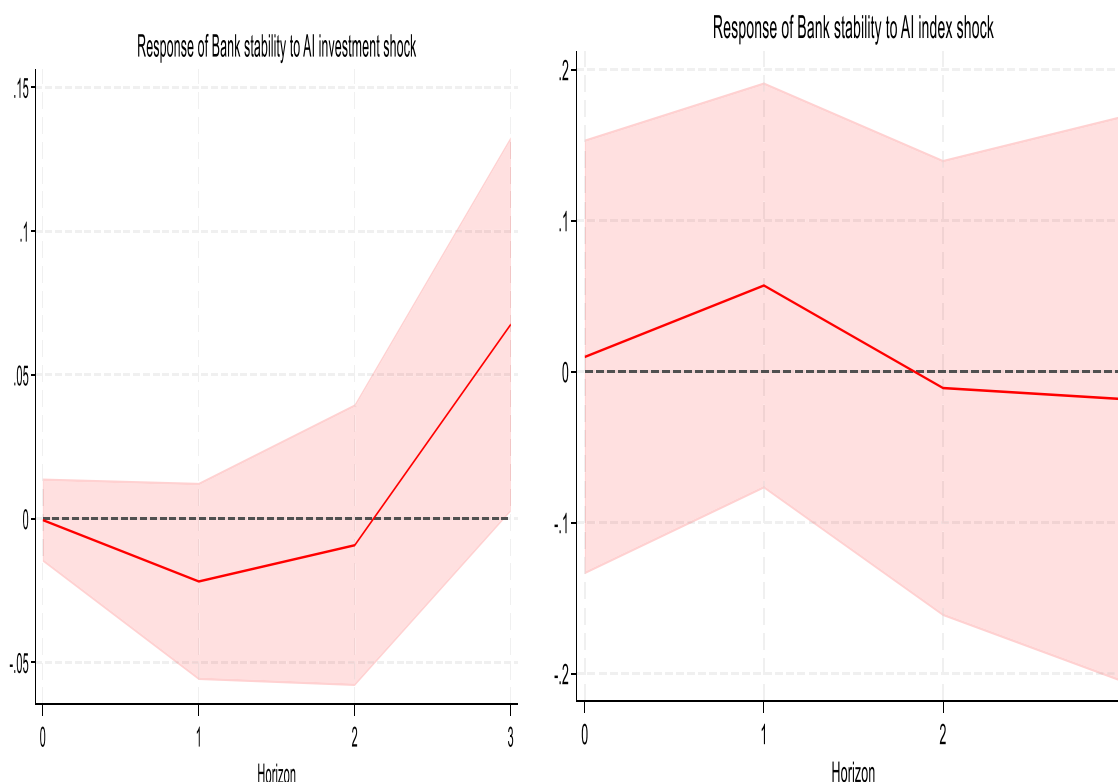


Fig. 3. Response of bank stability to artificial intelligence shock.

limitations, particularly due to the relatively short period of the data used for the analysis.

Policy recommendations from this analysis include several important actions. First, it is necessary to regulate inflation levels in the banking sector to strengthen its stability. Next, agricultural sector financing should be supported by integrating artificial intelligence technologies to better manage risks of uncertainty. This would help improve agricultural loan repayment, thus strengthening banking stability. Furthermore, investing in the development of artificial intelligence technologies that can help banks better manage payment defaults would further stabilize the banking system. The policies also aim to reduce payment defaults through increased use of artificial intelligence by financial institutions, with the goal of improving their ability to predict borrower risk and enhance banking stability. They also include promoting education on AI-related technologies for users of financial institutions, to facilitate both its adoption and the understanding of the results it generates. The definition of a clear and precise regulatory framework for artificial intelligence is essential to ensure transparency in its use, particularly in credit risk assessment and credit granting. Finally, financial authorities must ensure that banking regulation remains appropriate, adhering to the threshold identified in this study to maintain and strengthen banking stability in developed countries.

CRedit authorship contribution statement

Emile Sègbégnon Sonehekpon: Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Conflicts of interest

I know of no conflicts of interest associated with this publication, and there has been no significant financial support for this work that could have influenced its outcome. As Corresponding Author, I confirm that the manuscript has been read and approved for submission by me.

Data availability

Data will be made available on request.

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